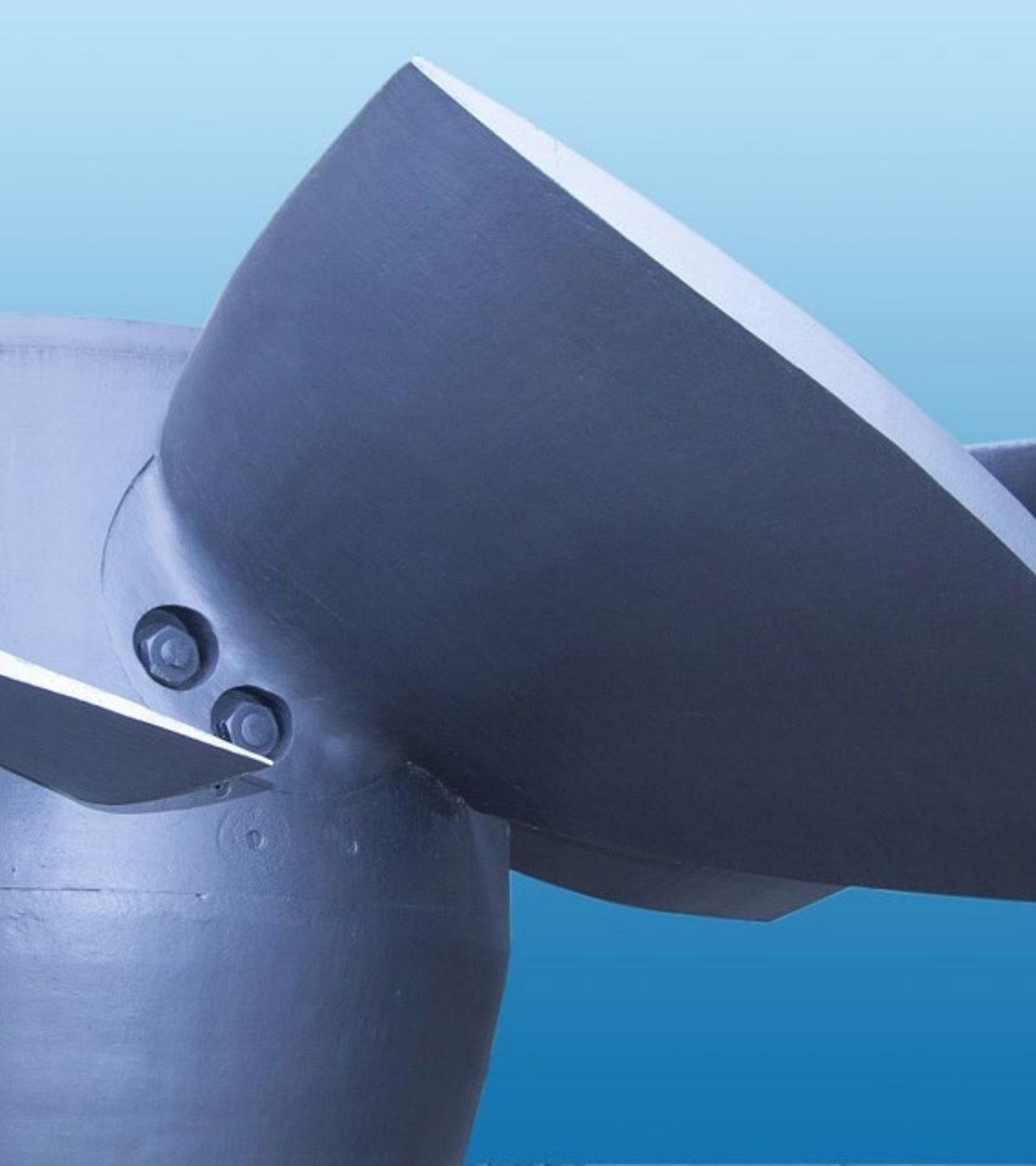




ME 300: Engineering Thermodynamics

Review of Devices

Review of devices

Passive ($\dot{W}=0$)

→ nozzles - $v \uparrow$ - $\Delta p_e = \dot{Q} = \dot{W} = 0$

→ diffusers - $v \downarrow$ - $\Delta p_e = \dot{Q} = \dot{W} = 0$

→ turbines - $P \downarrow$ - $\Delta k_e = \Delta p_e = \dot{Q} = \dot{W} = 0$

→ heat exchangers - $\dot{Q}$ - $\Delta k_e = \Delta p_e = \dot{W} = 0$

Work Input/output

→ compressor - work in - $\Delta k_e = \Delta p_e = \dot{Q} = 0$

→ turbine - work out - $\Delta k_e = \Delta p_e = \dot{Q} = 0$